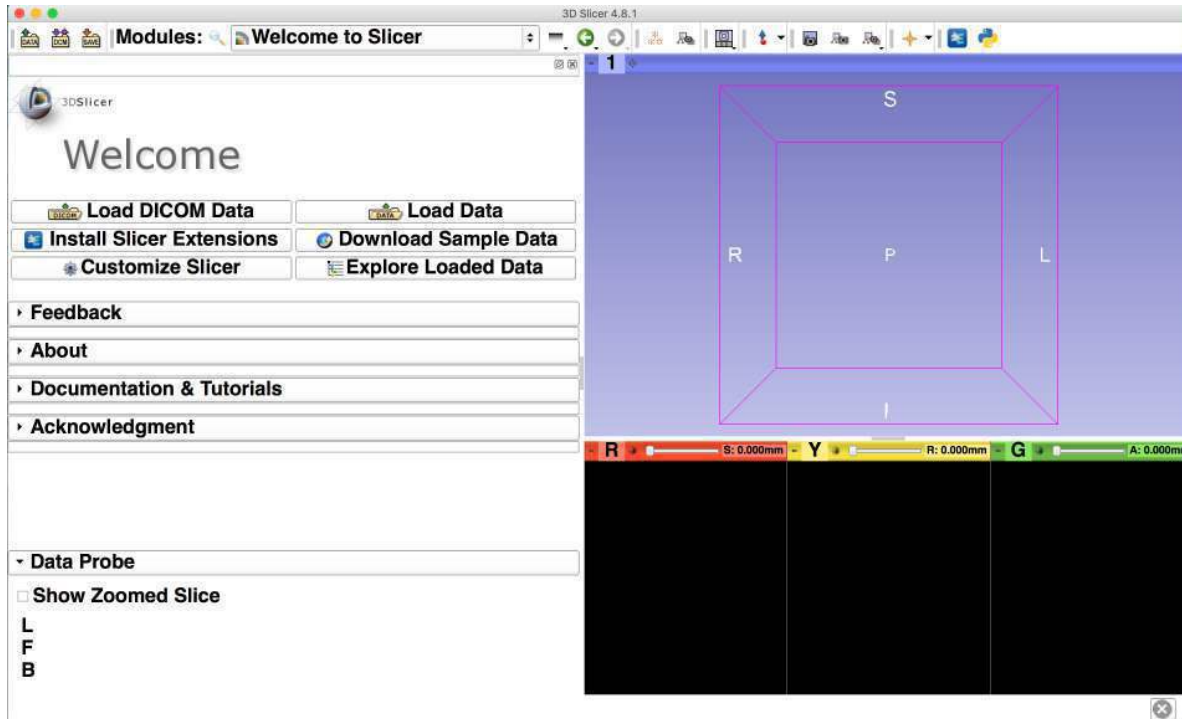


# Slicer Welcome

Sonia Pujol, Ph.D.

Assistant Professor of Radiology  
Brigham and Women's Hospital  
Harvard Medical School

# Goal

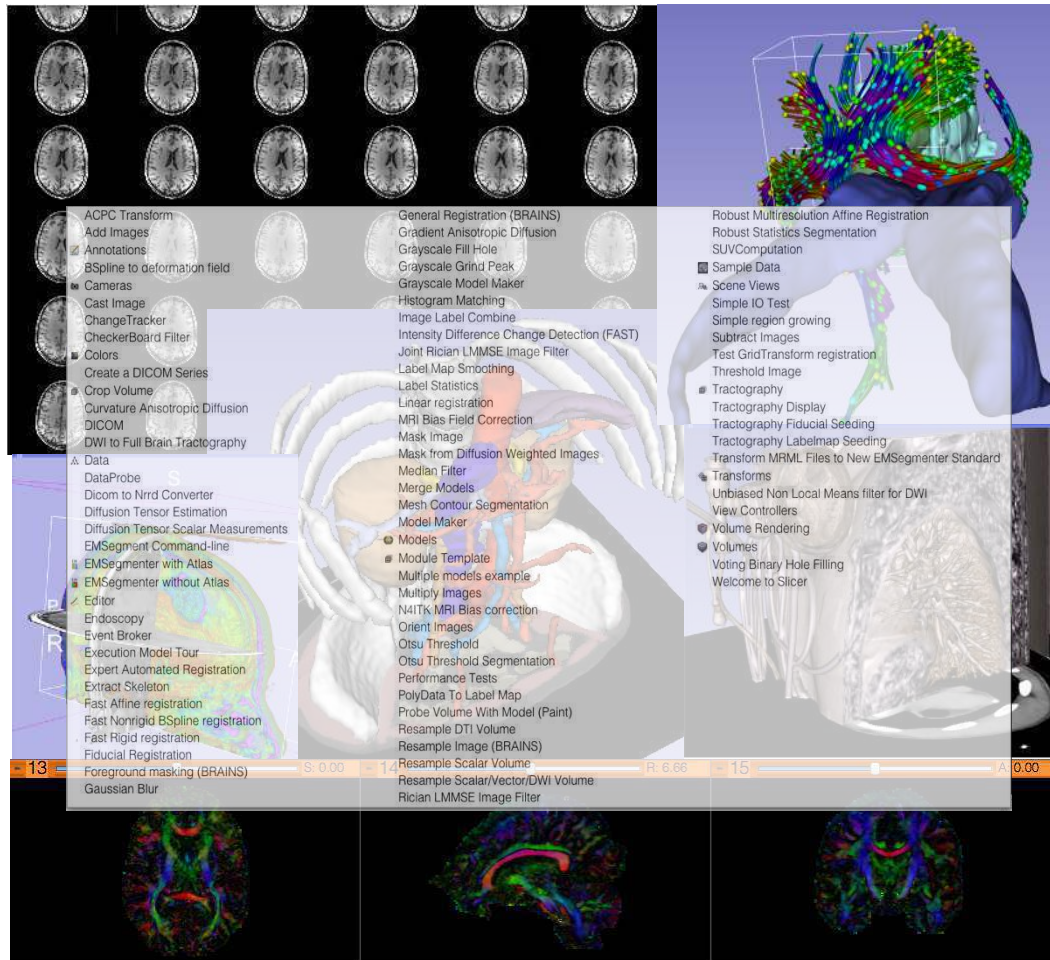


This tutorial is a short introduction to the Welcome module of the Slicer open-source software.

# Slicer4 Basics

- Slicer is an open-source software for segmentation, registration and visualization of medical imaging data
- The platform is developed through a multi-institution effort of several NIH funded large-scale consortia.
- Slicer is for medical research only, and is not FDA approved

# Slicer4 Basics



3D Slicer 4 version 4.8.1 includes 130 modules and 74 extensions for image segmentation, registration and 3D visualization of medical imaging data.

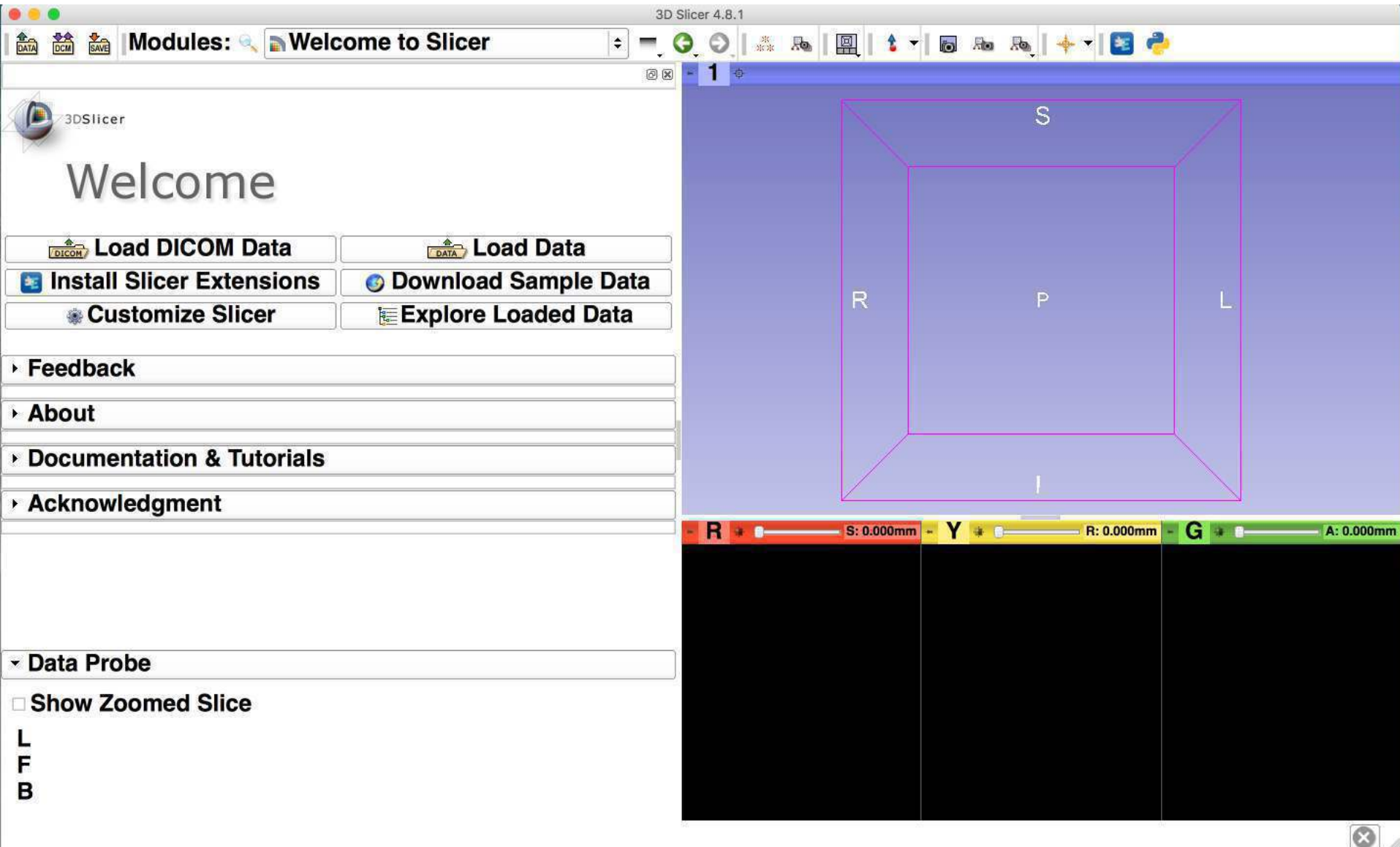
Images courtesy of Ron Kikinis, MD

Slicer Welcome - Sonia Pujol, Ph.D., NA-MIC  
ARR 2011-2018

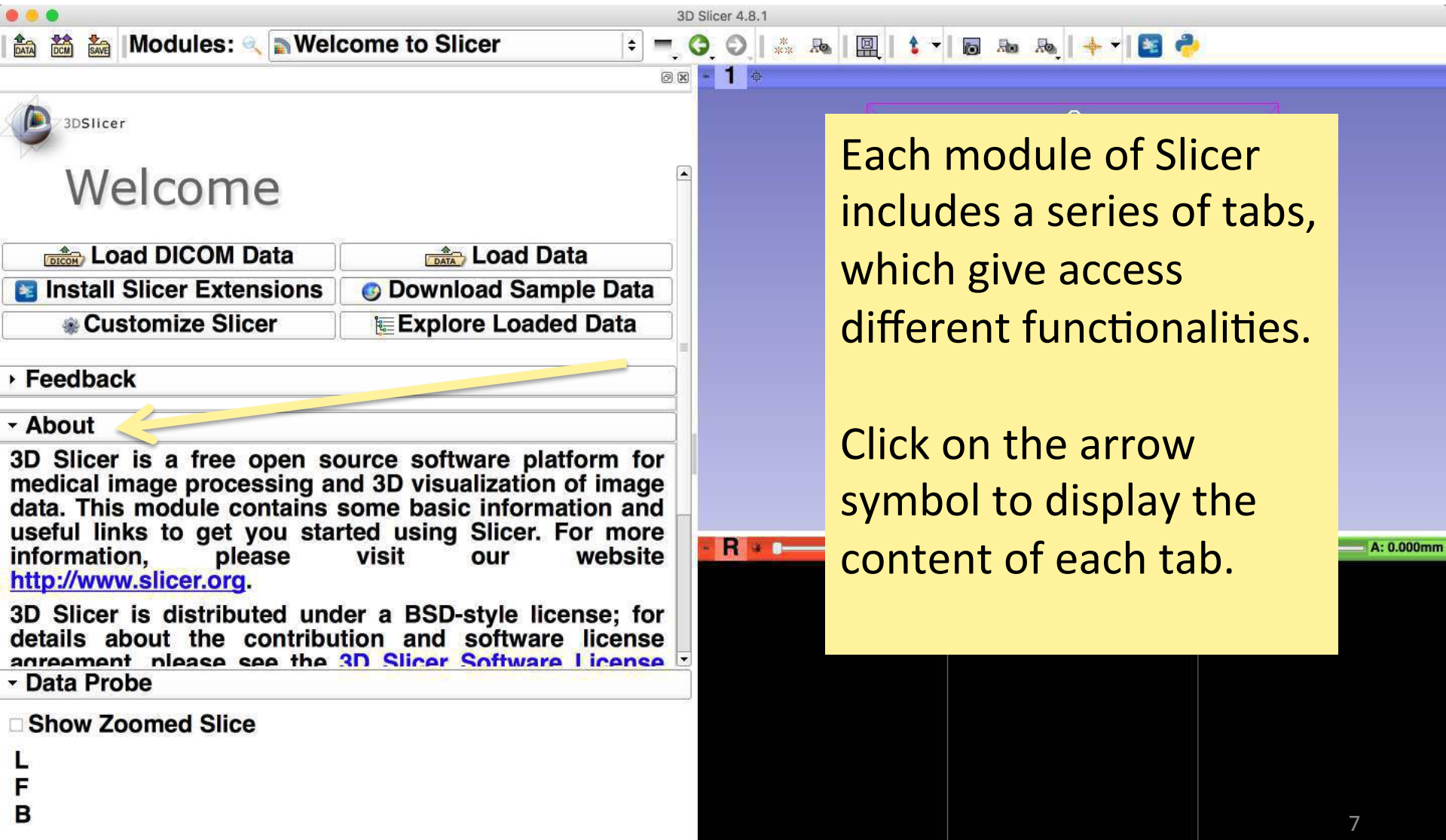
# Supported Platforms

- Slicer is a multi-platform software developed and maintained on Mac OSX, Linux and Windows.
- Slicer requires a minimum of 2 GB of RAM and a dedicated graphic accelerator with 64 MB of on-board graphic memory.

# 3D Slicer version 4.8



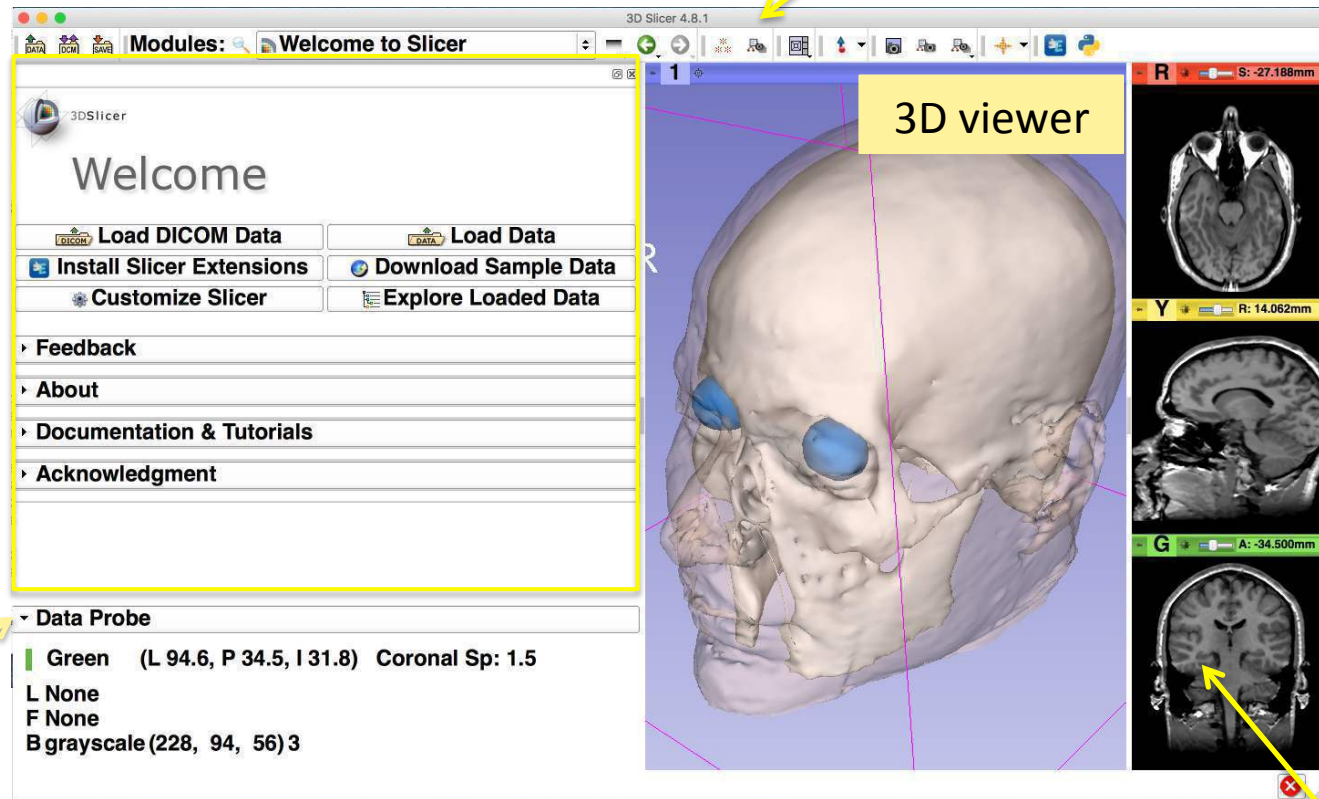
# Welcome to Slicer





# Slicer User Interface

Toolbar



User Interface (UI) panel of the Slicer Welcome Module

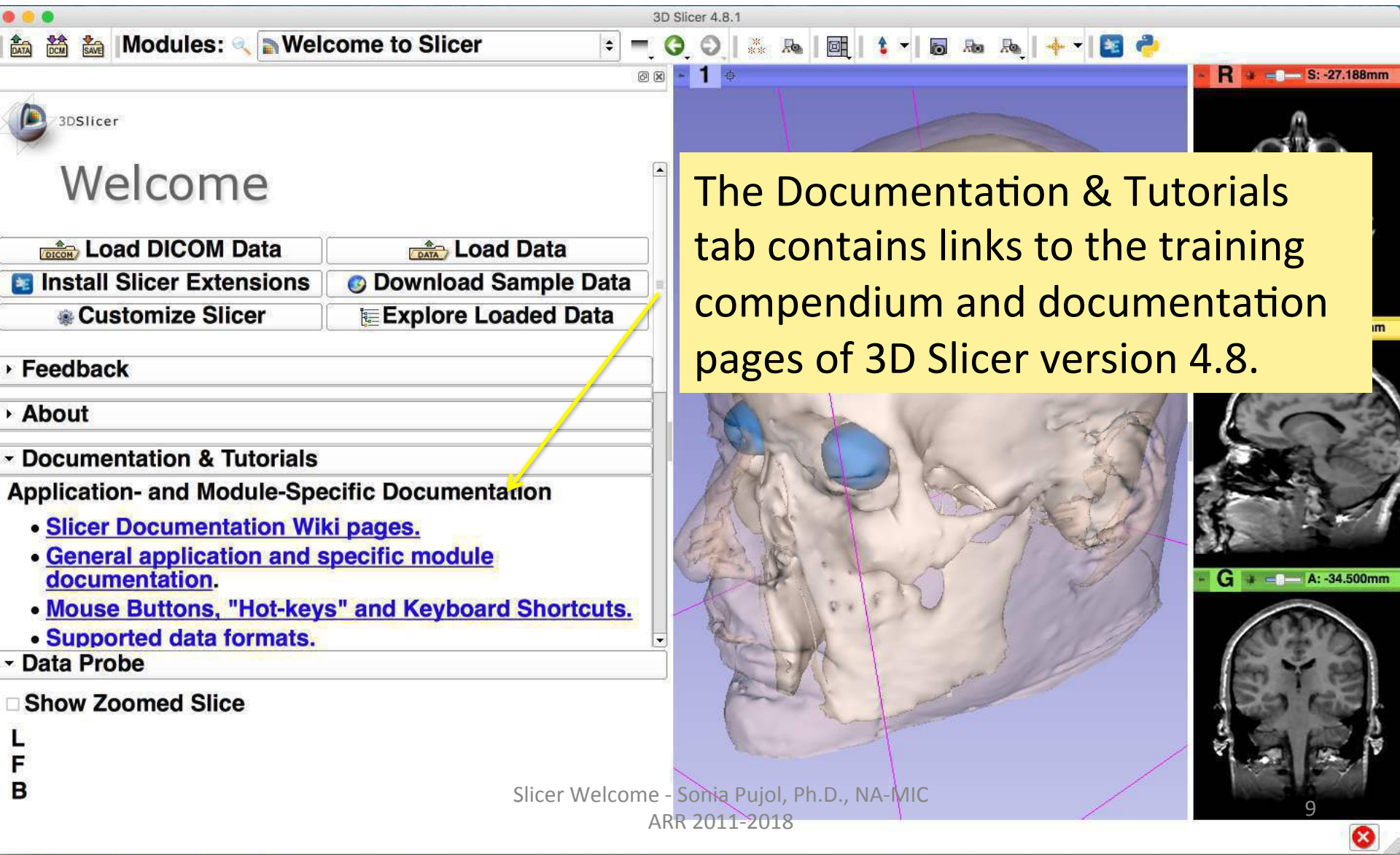
3D viewer

Data Probe

2D anatomical viewers



# Welcome Module



# Slicer 4.8 Training & Documentation

## Documentation/4.8/Training

[Home](#) < [Documentation](#) < 4.8 < [Training](#)

### Introduction: Slicer 4.8 Tutorials

- This page contains "How to" tutorials with matched sample data sets. They demonstrate how to release) to accomplish certain tasks.
  - For tutorials for other versions of Slicer, please visit the [Slicer training portal](#).
  - For "reference manual" style documentation, please visit the [Slicer 4.8 documentation page](#)
  - For questions related to the Slicer4 Training Compendium, please send an e-mail to [Sonia Pujoi](#)
- 
- Some of these tutorials are based on older releases of 3D Slicer. The concepts are still useful but features will be different in updated versions.

#### Contents [hide]

- 1 Introduction: Slicer 4.8 Tutorials
- 2 Quick Start Guide
  - 2.1 Downloading and Installing Slicer
- 3 General Introduction
  - 3.1 Slicer Welcome Tutorial
  - 3.2 Slicer4Minute Tutorial
- 4 3D Visualization
  - 4.1 Slicer4 Data Loading and 3D Visualization
  - 4.2 Slicer4 3D Visualization of DICOM Images for Radiology Applications
- 5 Programming
  - 5.1 Slicer4 Programming Tutorial
  - 5.2 Developing and contributing extensions for 3D Slicer
- 6 Segmentation
  - 6.1 Slicer4 Image Segmentation
- 7 Registration
  - 7.1 Slicer4 Image Registration
  - 7.2 Slicer Registration Case Library
- 8 Slicer Extensions
  - 8.1 Slicer4 Diffusion Tensor Imaging Tutorial
  - 8.2 Slicer4 Neurosurgical Planning Tutorial
  - 8.3 Slicer4 Quantitative Imaging tutorial
  - 8.4 Slicer4 IGT

## Documentation/4.8

[Home](#) < [Documentation](#) < 4.8

[Nightly](#) [4.8](#) [4.6](#) [4.5](#) [4.4](#) [4.3](#) [4.2](#) [4.1](#) [4.0](#) [3.6](#) [3.5](#) [3.4](#) [3.2](#) [ALL VERSIONS](#)

### Where to start ?

- [Getting started](#)  
Quick overview about Slicer
- [Training pages](#)  
Information on how to use Slicer 4.8
- [FAQ](#)  
Set of common questions/answers
- [Discussion Forum](#)   
The most effective way to get help from the community

### How to

- [Report a problem / Create a feature request](#) **NEW**

### Slicer Application

- [Installation / Uninstallation](#)
- [Main Application User-Interface](#)
- [Application Settings](#)
- [Extensions Manager](#)
- [Mouse Buttons, "Hot-keys" and Keyboard Shortcuts](#)
- [Recommended Computer Configurations](#)
- [Loading or Saving data and listing of supported data formats.](#)
- [Slicer Lookup Tables](#)

### Modules

-  [Annotations](#)
  -  [Data](#)
  -  [Data Store](#)
  -  [DICOM](#)
  -  [Editor](#)
  -  [Markups](#)
  -  [Models](#)
  -  [Scene Views](#)
  -  [Segmentations](#)
  -  [Segment Editor](#) 
  -  [Transforms](#)
  -  [View Controllers](#)
  -  [Volume Rendering](#)
  -  [Volumes](#)
  -  [Welcome to Slicer](#)
- [Modules by category](#)
  - [Modules by name](#)
  - [Modules by contributing organization](#)
  - [Modules by contributing individual](#)
  - [Modules by type](#)
  - [Modules by extension](#)
  - [Extensions by category](#)
  - [Extensions by name](#)
  - [Extensions by contributing organization](#)
  - [Extensions by contributing individual](#)

### Developers Corner

- [Information for Software Developers](#) **UPDATED**  
Source code, contribute patch, develop modules

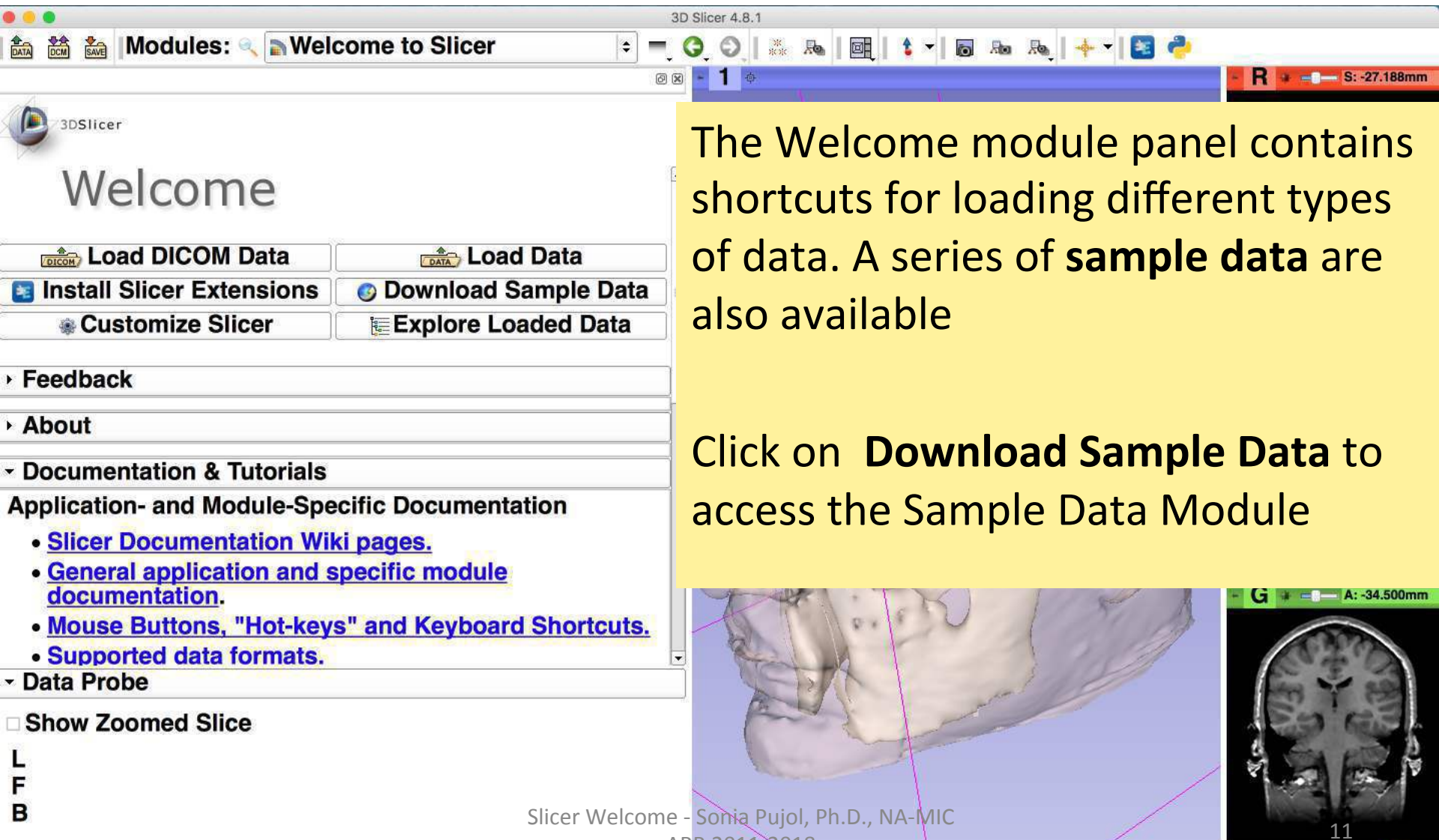
### Miscellaneous

- [Documentation guidelines](#)  
Slicer user documentation principle and guidelines
- [Visual blog](#)  
Set of screenshots showing Slicer in action.
- [Release Notes](#)  
Platform specific issues and considerations
- [Announcements & Acknowledgments](#)
- [Registration Library](#)   
Real-life example cases of using the Slicer registration tools, incl. datasets and step-by-step instructions to follow and try yourself.

### Documentation in other languages

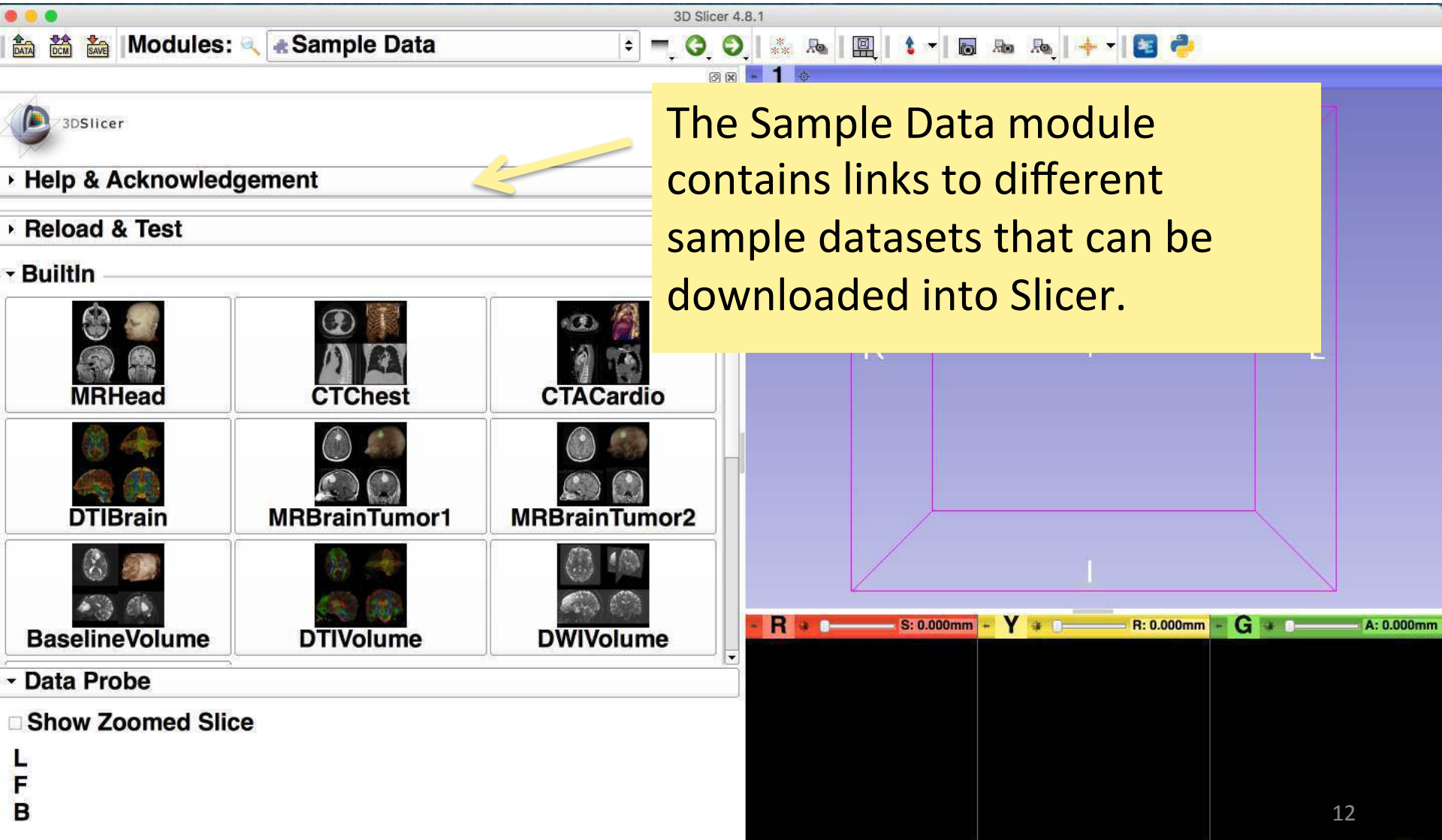
- [Español](#)

# Welcome Module

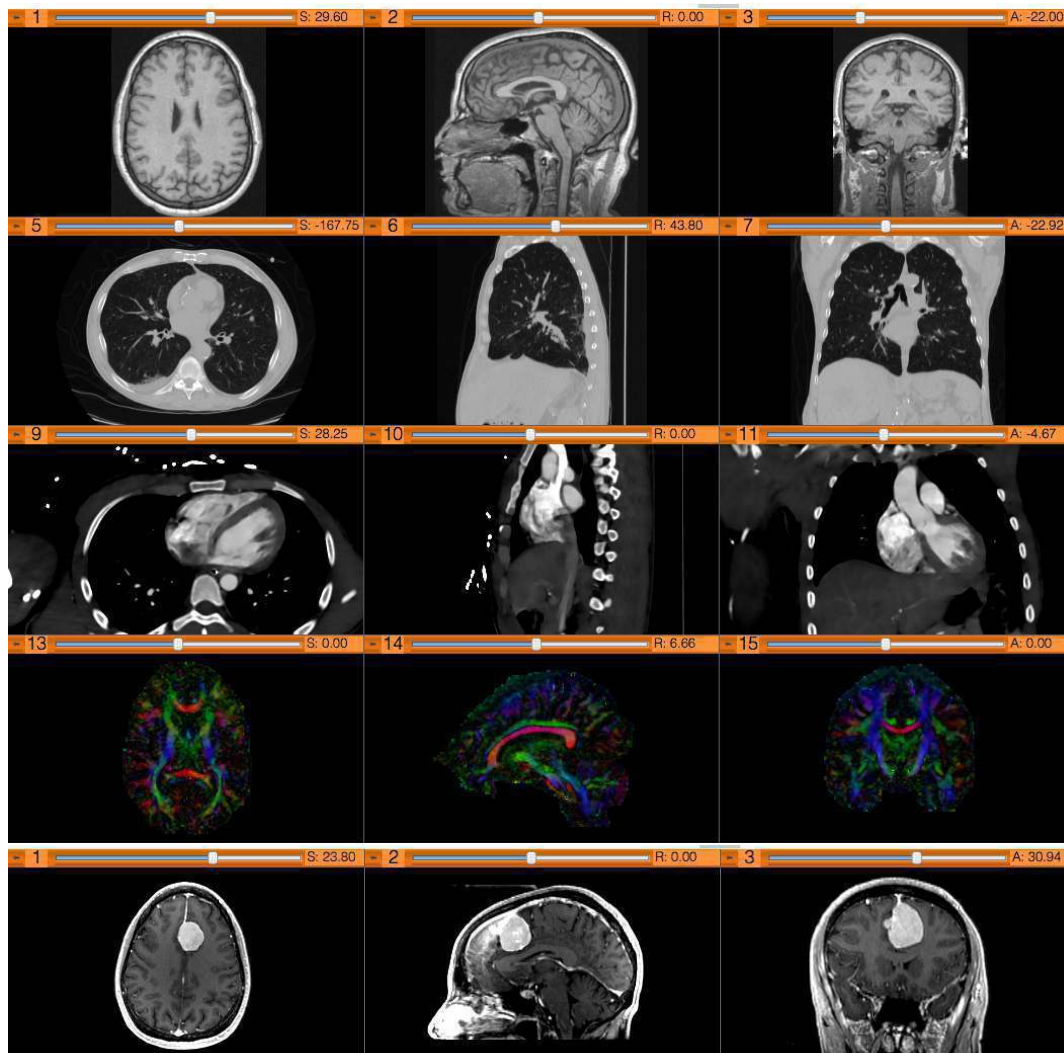




# Sample Data



# Sample Data



Brain MRI

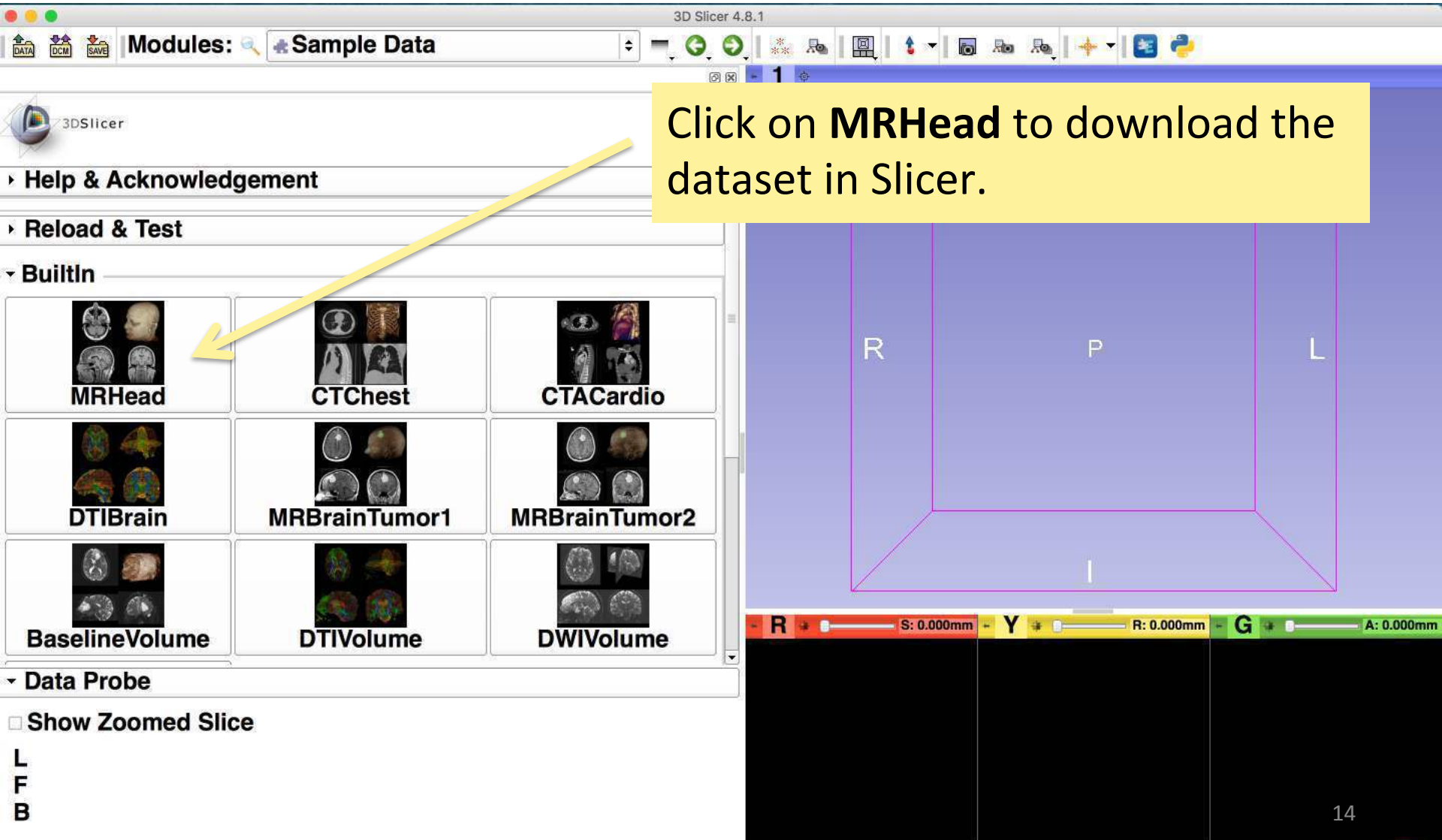
Chest CT

Cardiac CT

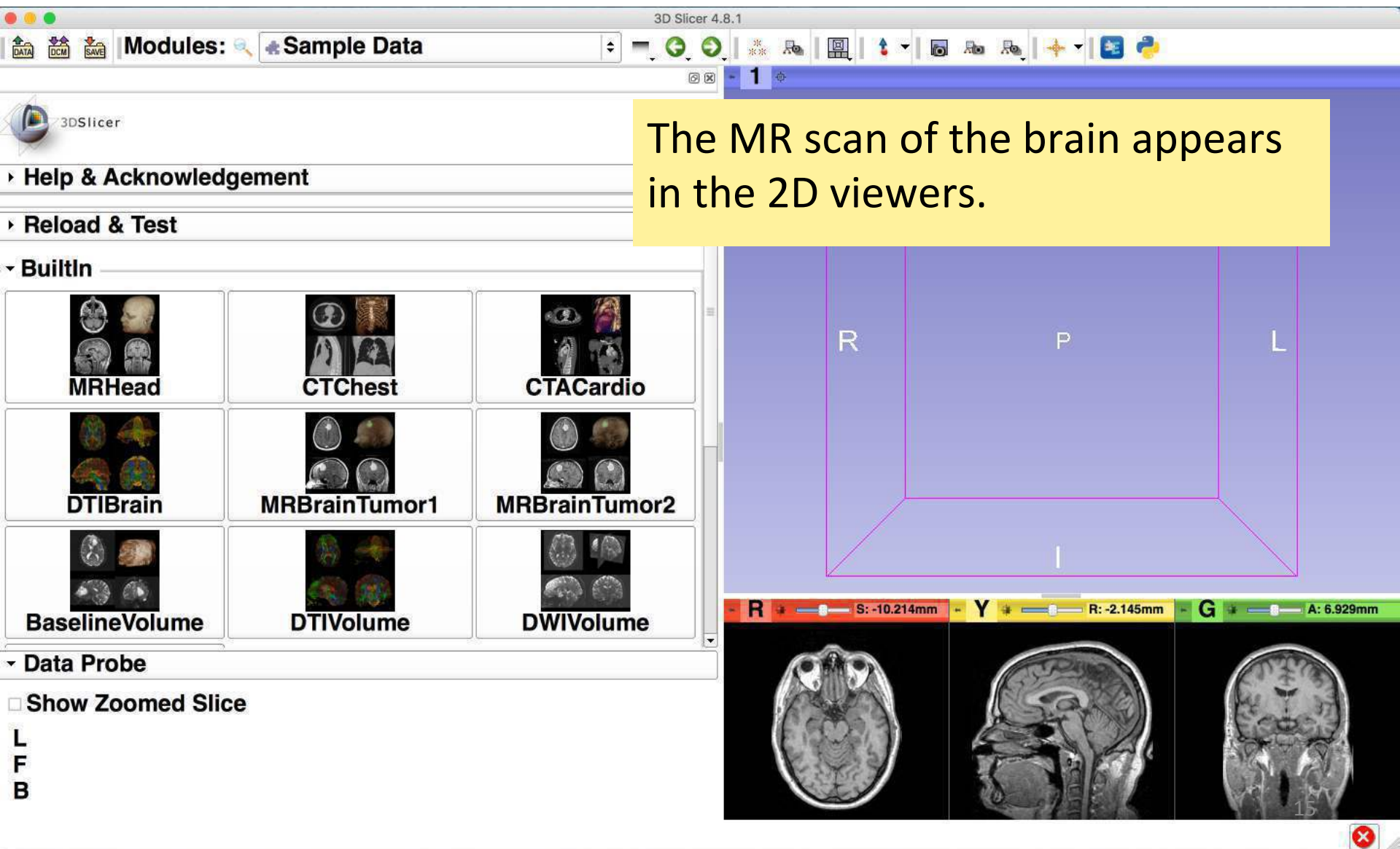
Diffusion Tensor  
Imaging (DTI) Dataset

Brain MRI  
(tumor patient)

# Sample Data

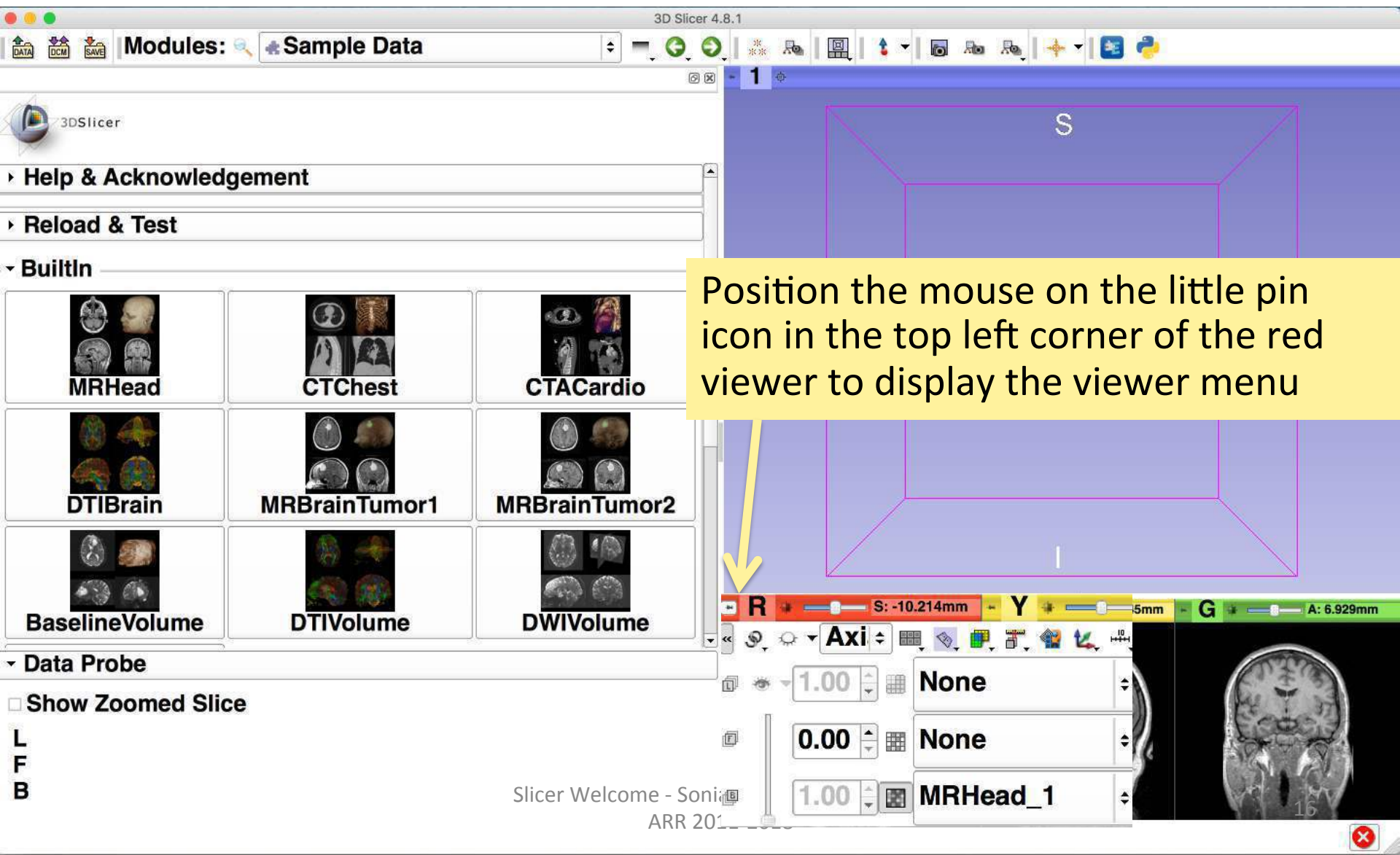


# Welcome Module

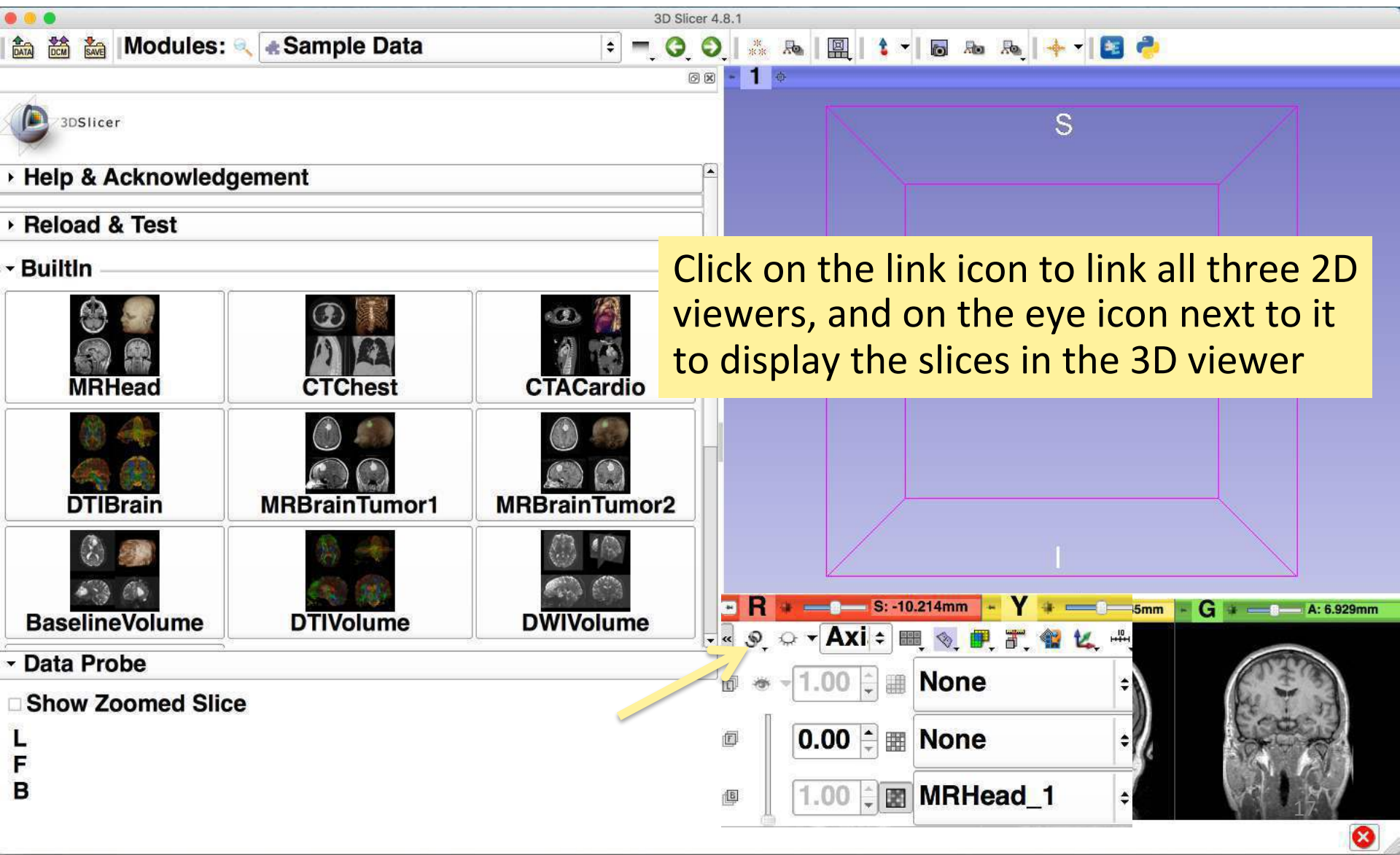




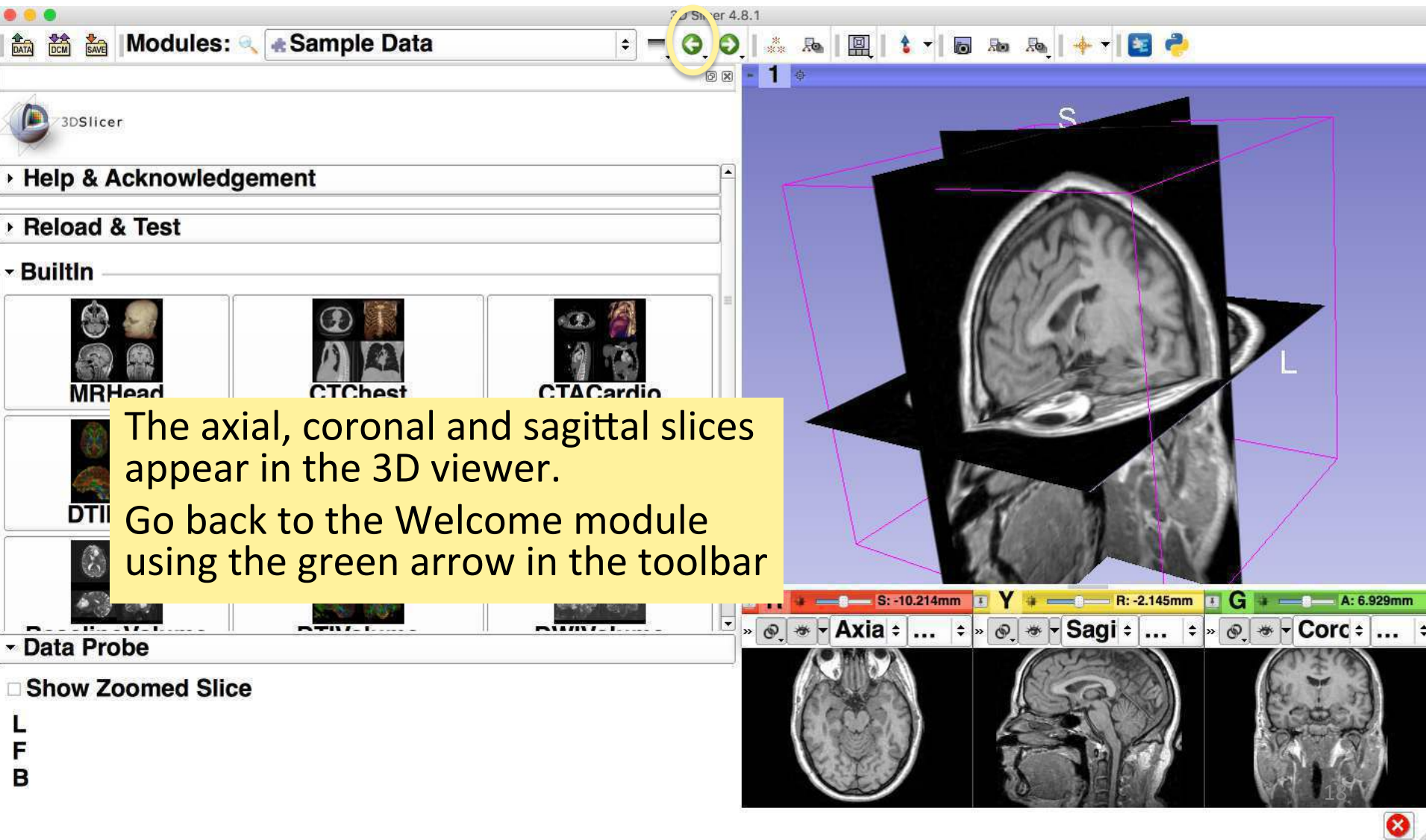
# MR Brain Sample Dataset



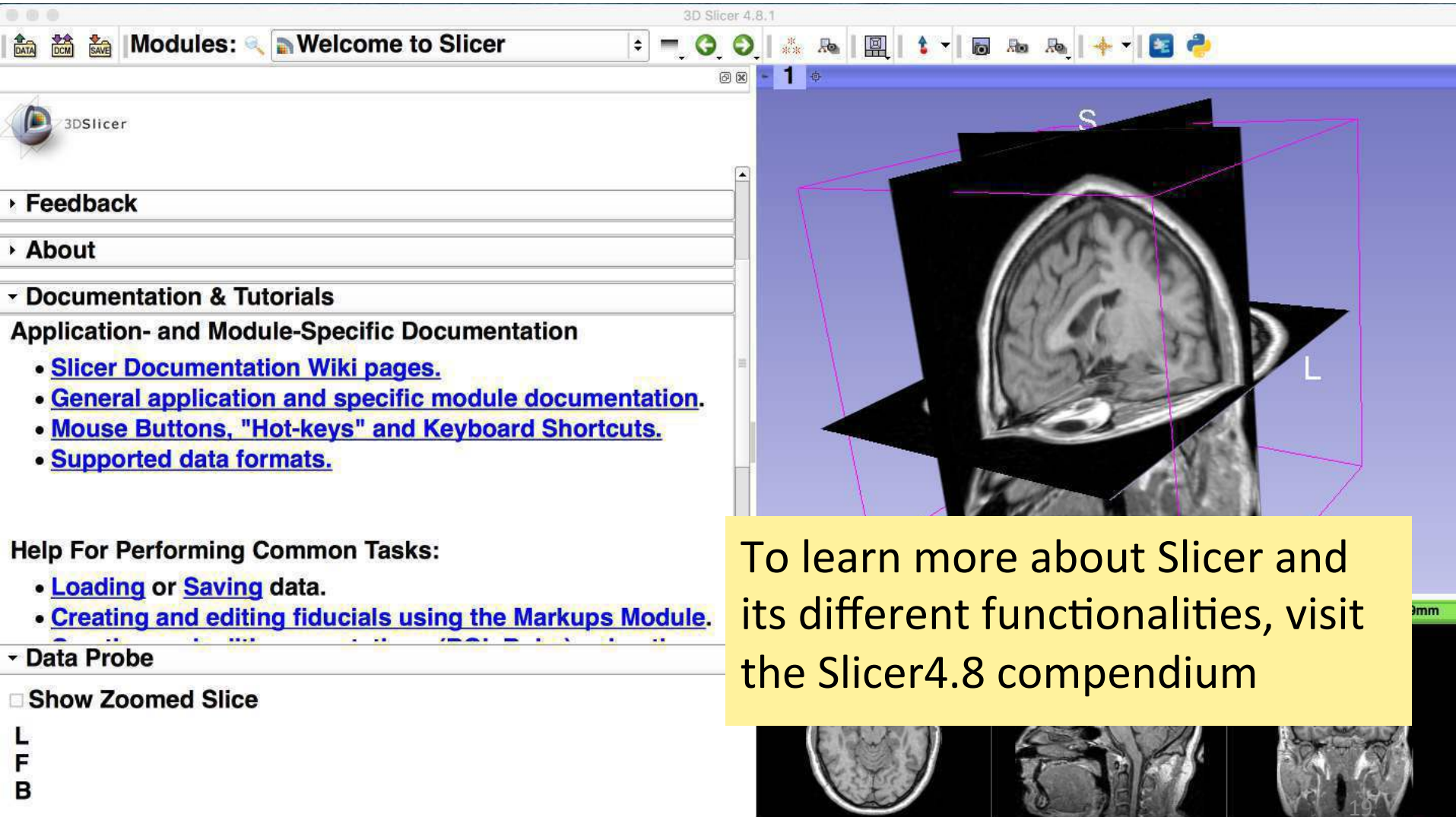
# MR Brain Sample Dataset



# MR Brain Sample Dataset

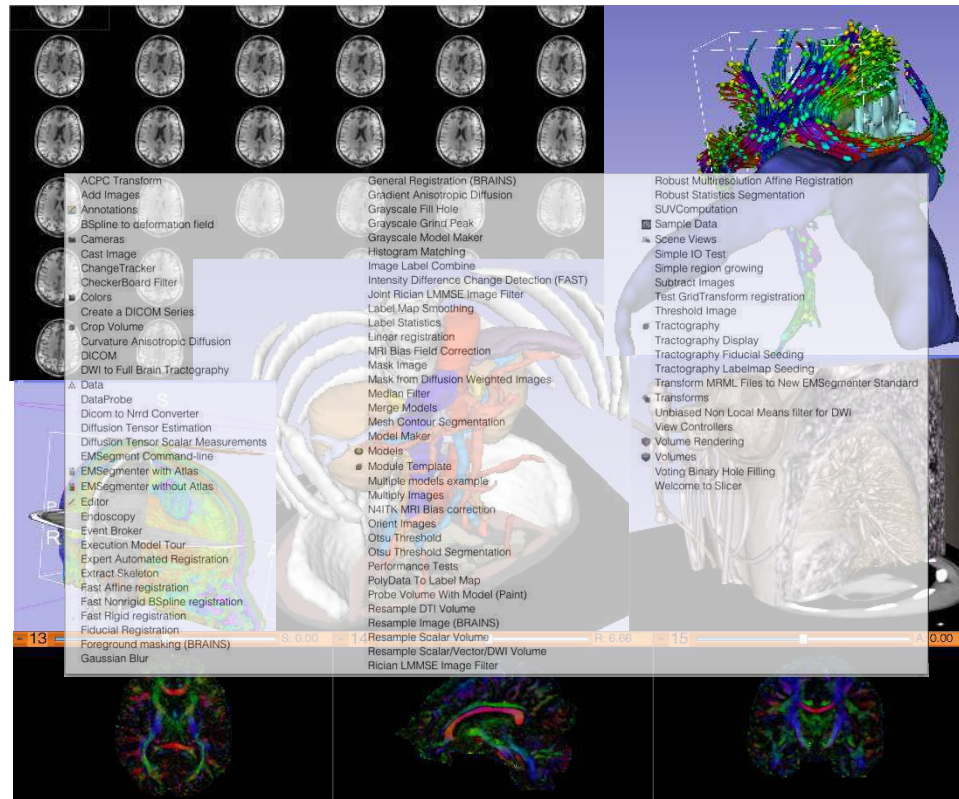


# Going Further





# Going Further



<http://www.slicer.org/slicerWiki/index.php/Documentation/4.8/Training>



# Acknowledgments



**National Alliance for Medical Image Computing**

NIH U54EB005149



**Neuroimage Analysis Center**

NIH P41EB015902